

Phase 2 Evaluation

Automated Resume Screening Using NLP: Evaluating Fairness and Mitigating Bias

Presented by: Yeshika, Bedant Singhania, Gaurav Sharma, Javed Akhtar, Manshvi Sharma

| Objectives & Problem Statement

The Problem

Manual resume screening is inefficient for large volumes and prone to human bias. Existing automated systems often inherit historical biases against specific demographics.

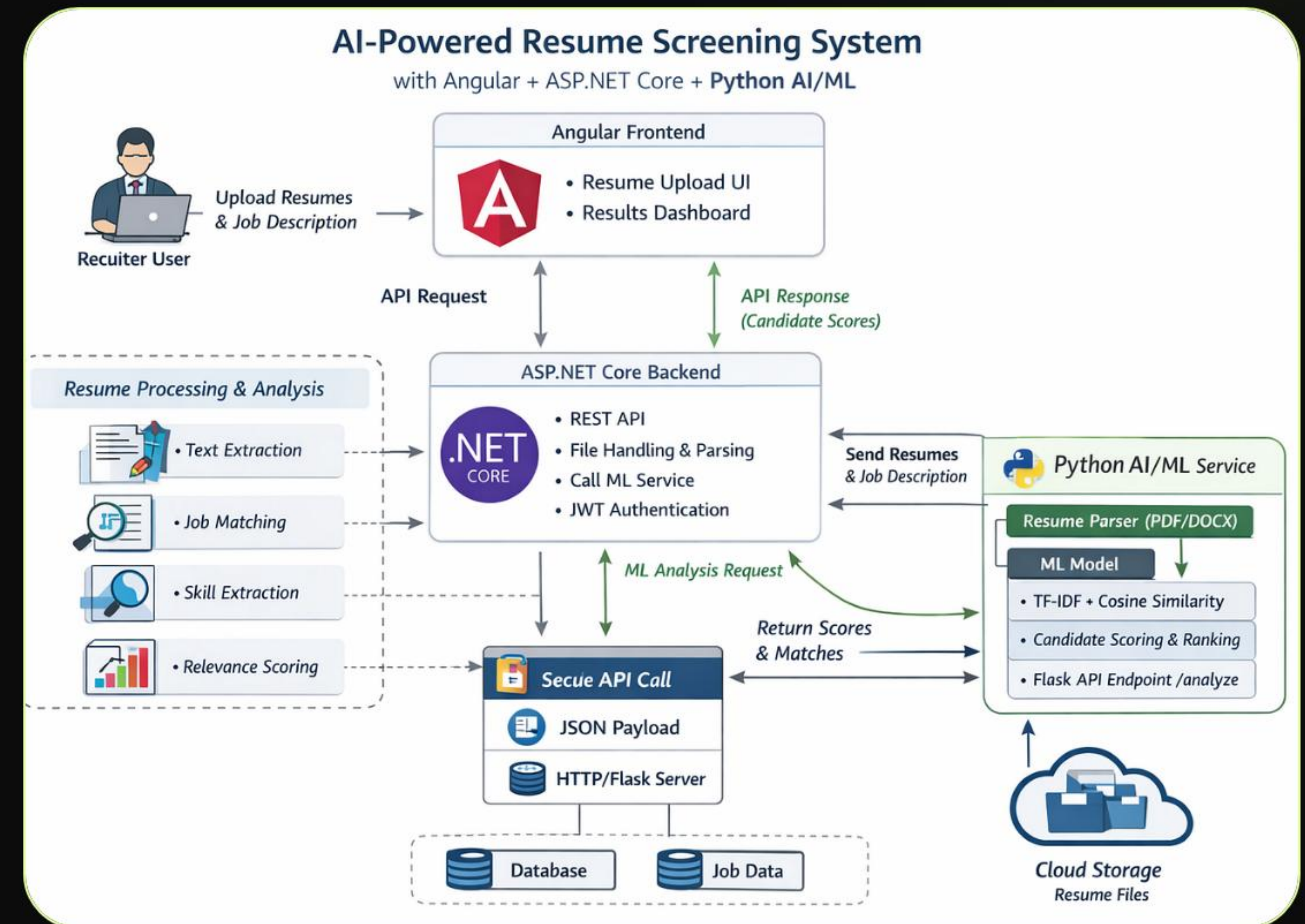
Key Objectives

Develop an NLP framework to classify resumes while detecting and mitigating demographic bias across gender, race, and age attributes.

| System Architecture Overview

The proposed framework follows a dual-track pipeline:

- **Pre-processing:** Tokenization & Lemmatization.
- **Feature Extraction:** TF-IDF and Transformer embeddings (BERT/RoBERTa).
- **Evaluation:** Monitoring fairness via DIR, DPD, and EOD metrics.
- **Mitigation:** Reweighting and adversarial debiasing.



| Literature Review Summary

Q What does it present?	The context of the study	A critical analysis of existing literature
Q What is its purpose?	To highlight the significance of the study	To evaluate the progress of knowledge in the research area
Q What does it do?	Introduces the topic in context and explains how it leads up	Helps identify the gaps in literature that led to the study

Evolution of ATS: Transitioned from keyword matching to deep contextual models.

Research Gap: Accuracy is often prioritized over fairness. Most studies lack integrated mitigation strategies during training.

Key Metrics: Identified DIR (Disparate Impact Ratio) as a primary compliance standard (0.80 threshold).

| Methodology & Technical Approach



Preprocessing

NLP cleaning, lemmatization, and tokenization using spaCy.



Model Training

LR, SVM, and RF for baselines; BERT & RoBERTa for context.



Fairness Loop

Continuous bias detection using fairness-aware optimization.

| Implementation Progress

1,200

Resumes Synthesized

5

Models Implemented

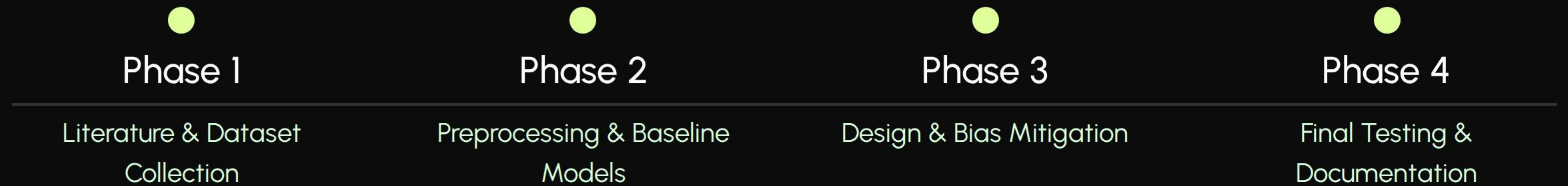
4

Demographic Groups

| Initial Performance Baseline

Model Architecture	Accuracy	F1-Score	Baseline DIR (Gender)
Logistic Regression + TF-IDF	82.3%	0.81	0.71
Support Vector Machine	83.7%	0.83	0.72
BERT (Fine-tuned)	91.7%	0.92	0.69

| Project Timeline & Milestones



How to create a data science project plan?

Scope & Boundaries

Focusing on software engineering roles, Evaluating gender, ethnicity, and age bias.
Offline dataset evaluation with 1,200 structured profiles.

Step 1

Design the
Problem
statement

Step 2

Sourcing
Data

Step 3

Exploratory
Data Analysis
& Data Cleaning

Step 4

Feature
Selection
& Engineering

Step 5

Visualize
your
Data

Step 6

Use
Predictive
Models

Step 7

Repeat,
Repeat,
Repeat!

"Good design is obvious. Great design is transparent and equitable."

— Project Thesis on Ethical AI

| Team Contributions

 Team Work

- **Yeshika & Bedant:** Model Fine-tuning.
- **Gaurav:** NLP Preprocessing Pipeline.
- **Javed:** Fairness Metric Implementation.
- **Manshvi:** Dataset Synthesis & Documentation.

Q&A

Thank you for your attention. We are ready for your questions.

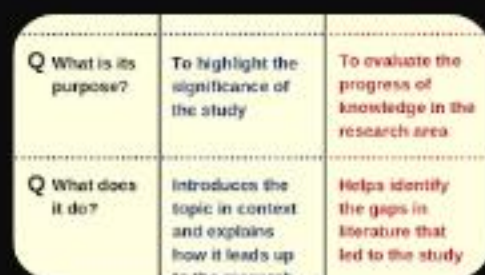
Chandigarh University | Department of CSE

| Image Sources



https://miro.medium.com/v2/resize:fit:1400/1*yEg7SuD_3FheEoBVHX8bQw.png

Source: medium.com



http://insights.cactusglobal.com/sites/default/files/6%20Differences%20between%20a%20study%20background%20and%20a%20literature%20review_0.png

Source: www.editage.com



https://images.stockcake.com/public/1/6/5/16572cd4-e2e2-4493-851c-7792d0474c2c_large/collaborative-planning-session-stockcake.jpg

Source: stockcake.com



https://daxg39y63pxwu.cloudfront.net/images/blog/data-science-project-plan/How_to_create_a_data_science_project_plan.webp

Source: www.projectpro.io